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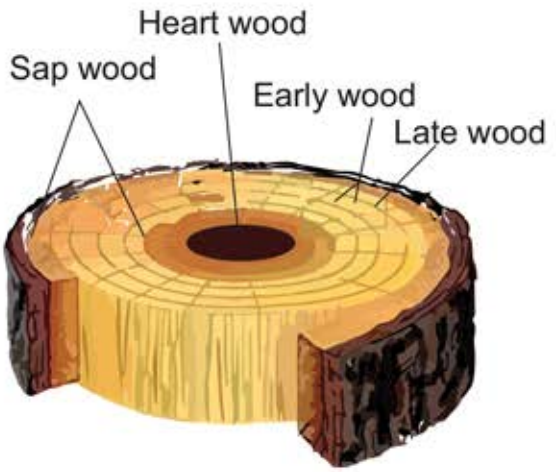


Figure 10.7: Structure of wood – Image shows early wood and late wood

Differences Between Spring Wood and Autumn Wood

Spring wood or Early wood	Autumn wood or Late wood
The activity of cambium is faster.	Activity of cambium is slower
Produces large number of xylem elements.	Produces fewer xylem elements.
Xylem vessels/ trachieds have wider lumen.	Xylem vessels/ trachieds have narrow lumen.
Wood is lighter in colour and has lower density	Wood is darker in colour and has a higher density.



The age of American, *Sequoiadendron* tree is about 3500 years.



Another feature of wood related to seasonal changes is the diffuse porous and ring porous condition. On the basis of diameter of xylem vessels, two main types of angiosperm woods are recognized.

❖ **Diffuse porous woods**

Diffuse porous woods are woods in which the vessels or pores are rather uniform in size and distribution throughout an annual ring.

Example: *Acer*

❖ **Ring porous woods**

The pores of the early wood are distinctly larger than those of the late wood. Thus rings of wide and narrow vessels occur.

Example: *Quercus*

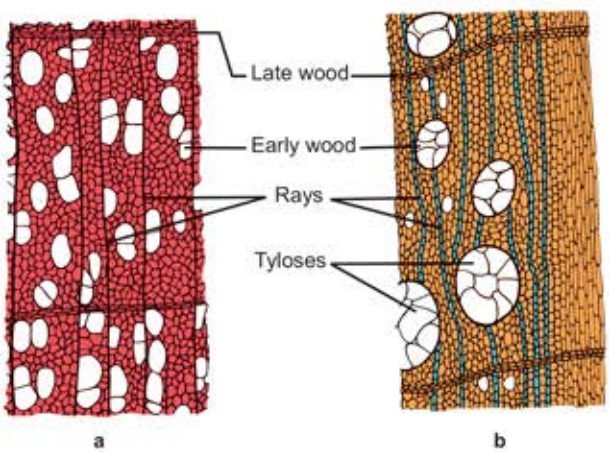


Figure 10.8: Transverse section of wood showing. a. Diffuse porous. b. Ring porous



The word "Porous" is used by the wood anatomists to refer to the appearance of the vessels as pores in transverse section.